

PART I

INTRODUCING OBJECT-ORIENTED PROGRAMMING

This part of the book introduces you to object-oriented programming. We'll discuss problems inherent in procedural code, then see how object-oriented programming addresses those concerns. Thinking in objects (with state and behavior) will give you a new perspective about how to write code.

Chapter 1 provides a review of procedural Python. I start by presenting a text-based card game named Higher or Lower, then work through a few progressively more complex implementations of a bank account in Python to help you better understand common problems with coding in a procedural style.

Chapter 2 shows how we might represent real-world objects in Python using classes. We'll write a program to simulate a light switch, modify it to include dimming capabilities, then move on to a more complicated TV remote simulation.

Chapter 3 gives you two different ways to think about what is going on behind the scenes when you create objects in Python.

Chapter 4 then demonstrates a standard way to handle multiple objects of the same type (for example, consider a simple game like checkers where you have to keep track of many similar game pieces). We'll expand the bank account programs from Chapter 1, and explore how to handle errors.